Blessed G/Michael Catholic Secondary School physics test one for grade 10

For each of the following questions, choose the best answer from the given alternatives.

A 1. A truck moves 70 m east, then moves 120 m west, and finally moves east again a distance of 90 m. What is the truck's resultant displacement?

A. 40 m, east B. 40 m, west C. 280 m, east D. 280 m, west

D 2. An object moves 20 m east in 30 s and then returns to its starting point taking an additional 50s. If west is chosen as the positive direction, what is the average ~~speed~~ velocity of the object?

A. 0.50 m/s east B. 0.50 m/s, west C. 0.73 m/s, east D. 0 m/s

B 3. If the acceleration is constant and negative, the motion is described as:

A. Speeding up B. Slowing down C. Non-uniformly accelerated D. At rest

C 4. A cart is given an initial velocity of 5.0 m/s and experiences a constant acceleration of 2.0 m/s^2. What is the magnitude of the cart's displacement during the first 6.0 s of its motion?

A. 10 m B. 55 m C. 66 m D. 80 m

B 5. A bird, accelerating from rest at a constant rate, experiences a displacement of 28 m in 11 s. What is its acceleration?

A. 0.21 m/s^2 B. 0.46 m/s^2 C. 0.64 m/s^2 D. 0.78 m/s^2

A 6. An automobile driver puts on the brakes and decelerates from 30.0 m/s to zero in 10.0 s. What distance does the car travel?

A. 150 m B. 196 m C. 336 m D. 392 m

D 7. A car travels 40 km at an average speed of 80km/h and then travels 40 km at an average speed of 40km/h. The average speed of the car for this 80-km trip is:

A. 40km/h B. 45km/h C. 48km/h D. 53km/h

B 8. In uniformly accelerated motion, which of the following remains constant?

A. Velocity B. Acceleration C. Displacement D. Speed

C 9. A car, initially at rest, travels 20 m in 4 s along a straight line with constant acceleration. The acceleration of the car is:

A A. 0.4m/s2 B. 1.3m/s2 C. 2.5m/s2 D. 4.9m/s2

B 10. A racing car traveling with constant acceleration increases its speed from 10m/s to 50m/s over a distance of 60m. How long does this take?

A. 2.0s B. 4.0s C. 5.0s D. 8.0s

C 11. A car moving at 12 m/s accelerates uniformly at 3 m/s². How much time does it take to reach 30 m/s?

A. 4 s B. 5 s C. 6 s D. 8 s

A 12. A car moving with an initial velocity of 25 m/s north has a constant acceleration of 3m/s2 south. After 6 seconds its velocity will be:

C A. 7 m/s north B. 7 m/s south C. 43 m/s north D. 20 m/s north

B 13. If the velocity of a body is zero at an instant, which of the following must also be zero?

A. Acceleration B. Displacement C. Speed D. None of the above

D 14. A body moving with a constant velocity of 15 m/s travels for 20 seconds. The distance covered is:

A. 150 m B. 200 m C. 250 m D. 300 m

A 15. An object with an initial velocity of 12 m/s west experiences a constant acceleration of 3 m/s2 west for 4 seconds. During this time the object travels a distance of:

A. 72 m B. 24 m C. 96 m D. 54 m

D 16. How far does a car travel in 6s if its initial velocity is 2 m/s and its acceleration is 2m/s2 in the forward direction?

A. 12 m B. 14 m C. 24 m D. 48 m

C 17. A freely falling body has a constant acceleration of 9.8 m/s2. This means that:

A. the body falls 9.8 m during each second

B. the body falls 9.8 m during the first second only

C. the speed of the body increases by 9.8 m/s during each second

D. the acceleration of the body increases by 9.8 m/s2 during each second

C 18. A particle starts from rest with acceleration 2 ~~m/s22\text{m/s}^22~~ m/s2. Its velocity after 8 seconds is:

A. 8 m/s B. 12 m/s C. 16 m/s D. 20 m/s

B 19. An object dropped from the window of a tall building hits the ground in 12.0 s. If its acceleration is 9.80 m/s2, the height of the window above the ground is:

A. 29.4m B. 58.8m C. 118 m D. 706 m

B 20. The area under a velocity-time graph represents:

A. acceleration C. speed

B. displacement D. change in velocity

B 21. A car moving at 12 m/s accelerates uniformly at 3 m/s². How ~~much time~~ far it travel does it take to reach 30 m/s?

A. 4 s B. 5 s C. 6 s D. 8 s

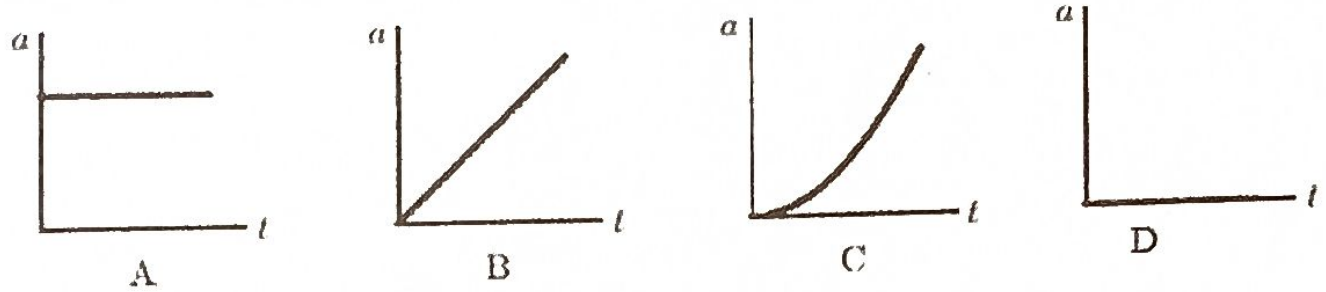
C 22. A bus starts from rest and attains a velocity of 30 m/s in 10 s under uniform acceleration. The distance traveled is:

A. 75 m B. 100 m C. 150 m D. 300 m

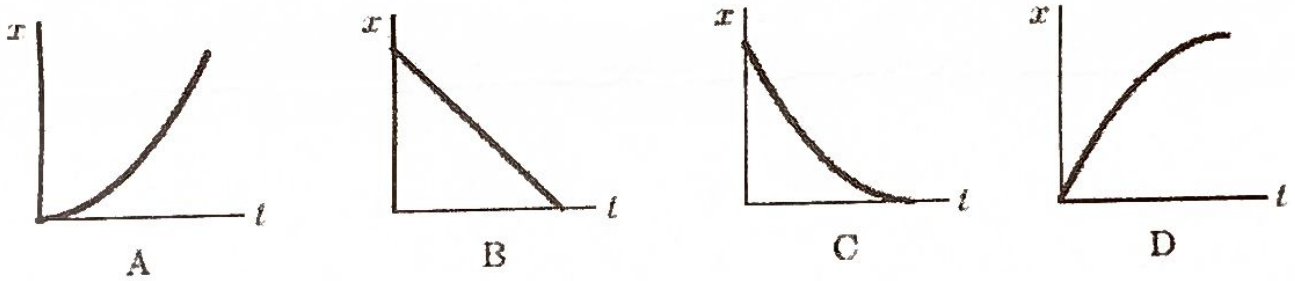
D 23. Displacement can be obtained from:

A. the slope of an acceleration-time graph
B. the slope of a velocity-time graph
C. the area under an acceleration-time graph
D. the area under a velocity-time graph

A 24. Which of the following four acceleration versus time graphs is correct for an object moving in a straight line with a uniformly increase in velocity ?



A 25. Which of the following four distances versus time graphs represents the motion of an object whose speed is increasing?



Name Bitanya Wondimagegnehu Grade and Sec. 10th B Number 12

1. A	11. C	21. B
2. D	12. A	22. C
3. B	13. BC	23. D
4. C	14. D	24. A
5. B	15. A	25. A
6. A	16. D	
7. D	17. C	
8. B	18. C	
9. C	19. BD	
10. BA	20. B	

Blessed G/Michael Catholic Secondary School

G-10 Physics Test **Topic: unit 1: Vectors** **Date: 25/02/2018 EC**

Time allowed: 1:30 hr

Name: Bitania Wondimagegnhu **Grade &sec:** 10th B **Number:** 12

I. **Write "True" if the statement is correct and "False" if it is incorrect.**

False 1. Two vectors with the same magnitude are necessarily equal. direction in dynamic

false 2. Two nonzero perpendicular vectors can be added together so their sum is zero.

True 3. To subtract vectors graphically, you add the negative of the vector being subtracted using the head-to-tail method. triangle

false 4. When two vectors are perpendicular, the magnitude of their resultant is simply the sum of their magnitudes. Pythagoras

false 5. Two vectors are said to be parallel vectors if and only if they are along the same direction and they have equal magnitude.

false 6. Three or more vectors with unequal magnitudes can be added together so their sum is zero.

false 7. The magnitude of the resultant of two vectors increase as the angle between them increases from 0° to 180°. A+B=C

True 8. If the magnitude of a vector has doubled, its direction remaining the same, then the magnitude of each component of the vector will be doubled.

II. **For each of the following questions, choose the best answer from the given alternatives.**

R 9. Which of the following is a scalar?

A. Velocity B. Temperature C. Force D. Displacement

A 10. What is the magnitude of the vector resulting from adding two vectors of magnitude 5 units each, pointing in exactly opposite directions?

A. 0 units B. 5 units C. 10 units D. 25 units

B 11. In a vector diagram, what does the arrowhead represent?

A. Magnitude B. Direction C. Origin D. Component

A 12. If a vector $\vec{v}$ has components V_x=6 and V_y = 8, what is the magnitude of $\vec{v}$?

A. 10 B. 14 C. 2 D. 48

C 13. Which of the following is **not** a vector?

A. Momentum B. Torque C. Electric current density D. Pressure

B 14. If vector $\vec{A}$ has components (6, 8) and vector $\vec{B}$ has components (2, 3), what are the components of $\vec{A}+\vec{B}$?

A. (4, 5) B. (8, 11) C. (6, 8) D. (12, 24)

$\cos 30° = sin60° = \frac{\sqrt{3}}{2}, cos37° = sin53° = 0.8, cos45° = sin45° = \frac{\sqrt{2}}{2}, cos53° = sin37° = 0.6, cos60° = sin30° = 0.5$

B 15. Which one of the following quantities is a vector?

A. The cost of a theater ticket.
B. The initial flight path from Bahir dar to Harar.
C. The current in a wire.
D. The population of the world.

A 16. A car moving east at 45 km/h turns and travels west at 30 km/h. What are the magnitude and direction of the change in velocity?

A. 15 km/h east
B. 15 km/h west
C. 75 km/h west
D. 75 km/h east

B 17. If $A^{\rightarrow}$=6 units east and $B^{\rightarrow}$=8 units north, what is the direction of the resultant vector $A^{\rightarrow}+B^{\rightarrow}$ measured from east?

A. 37° north of east
C. 37° east of north
B. 53° north of east
D. 53° east of north

A 18. A vector drawn 15 mm long represents a velocity of 30 m/s. How long should you draw a vector to represent a velocity of 20 m/s?

A. 10 mm
B. 20 mm
C. 5 mm
D. 20 mm

B 19. If two vectors have arrows of different lengths, what does that mean?

A. They have different directions
C. They have the same magnitude
B. They have different magnitudes
D. They are scalar quantities

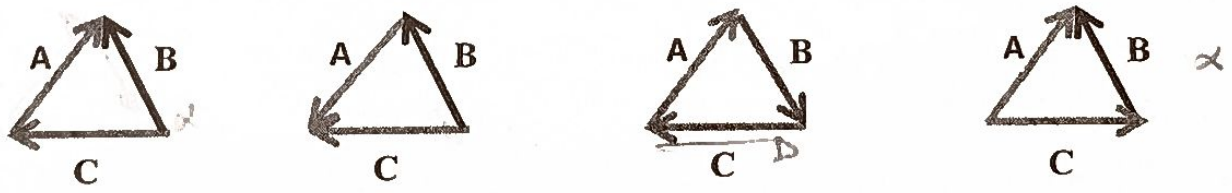
B 20. A vector that is 1 cm long represents a displacement of 5 km. How many kilometers are represented by a 3-cm vector drawn to the same scale?

A. 8 km
B. 15 km
C. 1.7 km
D. 0.6 km

B 21. What does the length of the arrow in a vector diagram represent?

A. Direction
C. Both magnitude and direction
B. Magnitude
D. Position

B 22. Which one of the vector diagrams (Figure 2.10) satisfies the vector equation **C = A + B**?



A.
B.
C.
D.

A 23. When two vectors are added graphically, the resultant vector's tail is placed at:

A. The tail of the first vector
C. The midpoint between the two vectors
B. The tail of the second vector
D. The head of the first vector

C 24. Two vectors have magnitudes of 7 and 13 units. But, their direction is unknown. What is the least magnitude of their resultant vector of their sum?

A. Zero
B. 20 units
C. 6 units
D. $\sqrt{659}$ units

B 25. A man is pushing his stalled car, trying to move it eastward. The force vector **F** that he applies to the car is 500 N, due east, what is the physical meanings of the vectors **-F**?

A. **-F** points due west and has a magnitude of -500 N.

B. **-F** points due west and has a magnitude of 500 N.

C. **-F** points due east and has a magnitude of -500 N.

D. **-F** points due east and has a magnitude of 500 N.

A 26. Vectors **A**, **B**, and **C** satisfy the vector equation **A**+**B** = **C**, and their magnitudes are related by the scalar equation A + B = C. How is vector **A** oriented with respect to vector **B**?

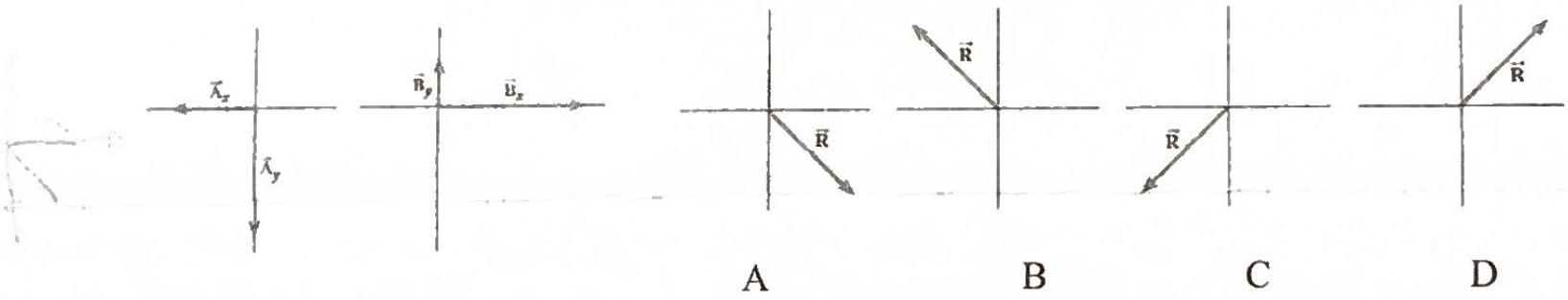
A. Vector **A** is parallel to vector **B** and they are along the same direction.

B. Vector **A** is parallel to vector **B** and they are in opposite direction.

C. Vector **A** is perpendicular to vector **B**, they are at 90° to each other.

D. Vector **A** is at an angle of 45° to vector **B**, they have equal magnitude.

A 27. Two vectors, **A** and **B**, have vector components that are shown (to the same scale) in the drawing. The resultant vector is labeled **R**. Which drawing shows the correct vector sum of **A** + **B**?



A B C D

B 28. Which of the following displacement vectors (if any) are equal?

A. Vectors **A** and **B**.

B. Vectors **A** and **D**.

C. Vectors **D** and **B**.

D. Vectors **C** and **B**.

Variable	Magnitude	Direction
$\vec{A}$	100 m	30° north of east
$\vec{B}$	100 m	30° south of west
$\vec{C}$	50 m	30° south of west
$\vec{D}$	100 m	60° east of north

III. Give short answer for each of the following questions.

1. Two vectors **A** and **B** are added together to give a resultant vector: **A** + **B** = **C**. The magnitudes of **A** and **B** are 3 m and 8 m, respectively, but the vectors can have any orientation. What are (a) the maximum possible value and (b) the minimum possible value for the magnitude of **C**?

a. max 11 m

b. min 5 m